INCH POUND
MIL-C-60235B
10 May 1989
SUPERSEDING
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15 April 1974 and
Amendment 1
16 October 1986

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, M91 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. *This specification covers* the requirements for the M91 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. *Unless otherwise specified, the issues of these documents are those* listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for
MIL-C-60539	Coating, Anodic, Conventional for Aluminum and Aluminum Alloy

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

5184883	Closed Bomb Assembly, M91 and CCU-79/A (6.365 cubic inch)

NAVAL ORDNANCE STATION (CAGE Code 19200 and 19203)

DL8597230	Cartridge, Impulse, M91 Assembly
8837884	Box, Fiberboard (Inner), Packing, Ammunition, For Cartridge, Impulse M91, M163, M169, M173, or M201
8837885	Box, Packing, Ammunition, For Cartridge, Impulse M91, M163, M169, M173, or M201
9227983	Box, Fiberboard (Outer), Alternative Packing Ammunition, For Cartridge, Impulse M91, M163, M169, M173, or M201

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Conformance to documents. The M91 Impulse Cartridge covered by this specification shall be manufactured in accordance with DL8597230 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the M29A1 Primer, Igniter Charge and the cartridge propellant charge are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

3.4.1.1 Primer. The primer shall conform to the applicable drawing and reference specification. Prior to assembly into a cartridge, the primer shall be visually examined in accordance with 4.6.2 and radiographically examined for defects as specified in 4.7.1.

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3.4.2 Hardness. The hardness of each component part shall be in accordance with the applicable drawing (see 4.7.2).

3.4.3 Protective finish. The finish of each component part shall be in accordance with MIL-C-60539 and the applicable drawing (see 4.7.3).

3.5 Assembly.

3.5.1 Radiographic examination. Each cartridge assembly shall be radiographically examined for defects (see 4.7.4).

3.5.2 Airtightness. Each assembled cartridge shall exhibit no air leakage greater than 1×10^{-4} standard cubic centimeters per second (see 4.7.5) at a 14.7 psia pressure differential.

3.5.3 Vibration. The assembled cartridge shall withstand vibration as prescribed in MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I—3.2.5 without evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component (see 4.7.6 and 6.2).

3.6 Functional requirements.

3.6.1 Primer sensitivity. The primer shall fire when drop tested at an energy level of 20 inch ounces in a sectioned cartridge void of igniter and propellant charge (see 4.7.7.1).

3.6.2 Cartridge performance.

3.6.2.1 Misfires. The cartridge shall not fail to fire.

3.6.2.2 Ballistics. The cartridge, when tested in a closed bomb (Drawing 5184883), shall produce the pressures and ignition delay at temperatures specified in table I (see 4.7.7.2).

TABLE I. *Ballistic requirements.*

Temperature (°F)	Delay time, max (ms)	Maximum pressure (psig)		Time to P-max (ms)
		Lower	Upper	
−60 to −70	65	1350	2050	80
65 to 75	65	1350	2050	80
195 to 205	65	1350	2050	80

3.7 Workmanship. Cartridges shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the supplier shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 30 assembled cartridges for ballistics testing including 12 for vibration tests, 10 sets of unassembled components for dimensional and visual examination, and a quantity in accordance with 4.4.2, i.e., cartridges assembled without the igniter and propellant charge for primer sensitivity test. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. When it becomes necessary to make a change in either the primer, propellant, or igniter lot after initial production, 12 assembled cartridges from the first production lot thereafter shall be tested in accordance with 4.7.6. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

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4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.4.2 Sample size for primer sensitivity. The number of cartridges assembled without igniter or propellant to be submitted for primer sensitivity tests shall be in accordance with table II.

TABLE II. *Primer sensitivity.*

Lot Size	Sample Size
3 to 15	3
16 to 25	4
26 to 40	5
41 to 65	7
66 to 110	10
111 to 180	15
181 to 300	25
301 to 500	30
501 to 800	35
801 to 1300	40
Over 1300	50

4.4.3 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.6.2.2. The sample for firing shall be conditioned to table III.

TABLE III. *Temperature conditioning.*

Temperature (°F)	Not vibrated	Vibrated
−60 to −70	6	4
65 to 75	6	4
195 to 205	6	4

4.4.4 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.5.1.2 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 and not greater than 3200.

4.5.2 Examination and tests. The quality conformance sample shall be subjected to the examinations of 4.6 and all the test procedures of 4.7 except for 4.7.6.

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4.6 Examination. All cartridges and components shall be examined visually and radiographically and characteristics defined in accordance with the applicable drawings. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and defect criteria specified in the contract (see 6.2 and 6.4). Four power (4×) magnification may be used in performing visual examination for such defects as burrs and foreign matter. All nonconforming items shall be rejected. When compliance is in doubt as a result of visual inspection, the characteristic shall be measured or gaged to determine acceptability. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

4.6.1 Cartridge, Impulse, M91 Assembly, Drawing 8597230.

Characteristic	Defect	Method of Inspection
Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Mixed ammunition	Visual
4	Metal defective	Visual
5	Marking missing, incorrect, or illegible	Visual
Major		
101	Cartridge profile failure requiring more than 20 pounds dead-weight to insert in concentricity receiver gage	Gage
102	Diameter of head, max	Gage
103	Diameter of body, max	Gage
104	Diameter of head recess, min	Gage
105	Length of datum diameter, max	Gage
106	Depth of face to head	Gage
107	Dent, failure to chamber in concentricity receiver gage with 20 pounds deadweight	Gage/Visual
108	Firing pin or retaining cup above flush of head face	Visual
109	Foreign matter	Visual
110	Burr	Visual
Minor		
201	Total length, max	Gage
202	Dent, which will chamber in concentricity receiver gage with 20 pounds deadweight, max	Gage/Visual
203	Draw scratch	Visual
204	Fold, wrinkle, buckle, or bulge	Visual
205	Varnish inadequate or missing	Visual
206	Surface finish improper	Visual

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4.6.2 Primer, Percussion, M29A1, Drawing 8798312, covering a detail of 8597230.

Characteristic	Defect	Method of Inspection
Critical		
1	Improper assembly	Visual/X-ray
2	Any part missing	Visual/X-ray
3	Primer pellet broken, cracked, or dislodged	Visual
4	Metal defective	Visual
5	Primer foil discolored	Visual
6	Mixed primers	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual

4.6.3 Case, Cartridge, Drawing 8597231, covering a detail of Drawing 8597230.

Critical		
1	Metal defective	Visual
Major		
101	Concentricity of large diameter with small diameter	Gage
102	Diameter of head, min	Gage
103	Thickness of base	Gage
104	Diameter of body, min	Gage
105	Thickness of wall	Gage
106	Mouth of radius missing	Visual
107	Protective finish inadequate or defective	Visual
108	Foreign matter	Visual
109	Burr or sharp corner	Visual
Minor		
201	Small diameter	Gage
202	Large diameter	Gage
203	Total length	Gage
204	Location of datum diameter	Gage
205	Flatness of base	Gage
206	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
207	Surface finish improper	Visual

4.6.4 Cup, Retaining, Drawing 8597232, covering a detail of Drawing 8597230.

Critical

None defined

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Characteristic	Defect	Method of Inspection
Major		
101	Concentricity of counterbore and outside diameter with bore	Gage
102	Diameter of bore	Gage
103	Diameter of counterbore	Gage
104	Outside diameter	Gage
105	Depth of counterbore	Gage
106	Total length	Gage
107	Metal defective	Visual
108	Surface finish of bore and outside diameter	Visual
109	Burr or sharp edges	Visual
Minor		
201	Foreign matter	Visual
202	Surface finish improper, except as otherwise classified	Visual

4.6.5 Head, Cartridge, Drawing 8597233, covering a detail of Drawing 8597230.

Critical		
None defined		
Major		
101	Concentricity of primer flash hole counterbore, large diameter of body, firing pin cavity, large face diameter, face counterbore, and O-ring seat with primer cavity	Gage
102	Diameter of primer flash counterbore	Gage
103	Diameter of primer pocket	Gage
104	Diameter of retaining cup cavity	Gage
105	Large diameter of face	Gage
106	Diameter of flange	Gage
107	Diameter of face counterbore	Gage
108	Length to O-ring seat, max	Gage
109	Length to datum diameter	Gage
110	Depth of face counterbore	Gage
111	Depth of primer pocket	Gage
112	Depth from face counterbore to primer pocket	Gage
113	Thickness of flash web	Gage
114	Radii of chamfers missing or incorrect	Visual
115	Metal defective	Visual
116	Surface finish improper in retaining cup cavity and primer pocket	Visual
117	Protective finish inadequate or defective	Visual
118	Foreign matter	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Small diameter of base	Gage
202	Large diameter of base	Gage
203	Length to O-ring seat, min	Gage
204	Total length	Gage
205	Length from face to base shoulder	Gage
206	Surface finish improper, except as otherwise classified	Visual
207	Burr, except where permitted	Visual

4.6.6 Plug, Firing, Drawing 8597234, covering a detail of Drawing 8597230.

Critical		
None defined		
Major		
101	Concentricity of body with flange	Gage
102	Diameter of body, min	Gage
103	Diameter of flange	Gage
104	Height of boss, min	Gage
105	Profile and position of striker	Gage
106	Metal defective	Visual
107	Burr or sharp edges	Visual
108	Surface finish of body improper	Visual
Minor		
201	Diameter of body, max	Gage
202	Total length	Gage
203	Length of body	Gage
204	Surface finish improper except as otherwise classified	Visual
205	Foreign matter	Visual

4.6.7 Ring, Retaining, Drawing 8597235, covering a detail of Drawing 8597230.

Critical		
None defined		
Major		
101	Concentricity of inside diameter with outside diameter	Gage
102	Outside diameter, min	Gage
103	Inside diameter, max	Gage
104	Height	Gage
105	Metal defective	Visual
106	Radius incorrect	Visual
107	Protective finish inadequate or defective	Visual
108	Burr	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Outside diameter, max	Gage
202	Inside diameter, min	Gage
203	Surface finish improper	Visual
204	Foreign matter	Visual

4.6.8 Box, Fiberboard (Inner), Ammunition, Packing, Drawing 8837884.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Separator or filler missing	Visual
203	Heat sealer incomplete, loose, or distorted	Visual
204	Cut or damaged through impregnated box	Visual
205	Marking missing, incorrect, or illegible	Visual

4.6.9 Box, Ammunition, Packing, Drawing 8837885.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly assembled	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual
207	Number of cartons incorrect	Visual

4.6.10 Box, Fiberboard (Outer), Alternative, Drawing 9227983.

Critical		
None defined		
Major		
None defined		

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Characteristic	Defect	Method of Inspection
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.7 Test methods and procedures.

4.7.1 *Primer x-ray.* *Prior to assembly into the cartridge, the primer* shall be radiographically examined in accordance with MIL-STD-453. Each primer shall be x-rayed on boards or trays with the cup side perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Primers are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified, shall be forwarded to the engineering activity designated by the acquiring activity. Interpretation of the radiographic films shall verify the following:

Presence of Primer Charge
No Material Defects

4.7.2 Hardness. Ten units from each heat-treated lot of components shall be tested with a hardness tester. Failure of any unit to comply with the requirements of the drawings shall be cause for rejection of the lot of components.

4.7.3 Protective finish. Unless otherwise specified (see 6.2 and 6.4), sampling plans and procedures in the determination of acceptability of coated parts submitted by the contractor shall be in accordance with the provisions set forth in tables IV and V and MIL-C-60539. The anodized part shall be sectioned longitudinally to better expose the critical surfaces to salt spray test.

4.7.4 Radiographic examination. All assembled cartridges shall be radiographically examined in accordance with MIL-STD-453. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number as stamped on the cartridges and the span of serial numbers displayed. Cartridges having any observable imperfections in assembly shall be cause for rejection. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Discontinuities in serial numbers shall be identified on the x-ray negative. Defective cartridges, found by the vendor, shall be clearly identified on the negative and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridge with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings (see 6.2).

4.7.5 Airtightness. Each cartridge shall be subjected to specified pressure differential. The method for determining airtightness shall be with a helium mass spectrometer. The helium leak rate shall be converted to standard cubic inches air. Failure to comply with the requirements of 3.3.2 shall be cause for rejection of the cartridge.

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4.7.6 Vibration test. Twelve of the 30 assembled cartridges of the first article sample, x-rayed as directed in 4.7.4, properly identified, shall be subjected to vibration. The method and procedures for performing the vibration test shall be in accordance with MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 (see 6.2). Then the 12 vibrated cartridges shall be x-rayed again. Evidence of any disintegration of the primer, propellant, or igniter, or derangement of any component shall be cause for rejection of the sample.

4.7.7 Functional. The following tests shall be performed by the Government.

4.7.7.1 Primer sensitivity. A quantity of cartridge heads, in accordance with table II, void of igniter and propellant, from the first article sample and each subsequent lot shall be tested for sensitivity. In the event loaded cartridges are selected for test, remove and discard all powder charges from the test cartridge by sectioning the cartridge case around the periphery, 1/2 inch maximum above the base. Drop test at 10 inches (adjusted) using a steel ball 15/16-inch in diameter, weighing 1.94 ± 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the acquiring agency. The height of 10 inches shall be adjusted taking into consideration the actual weight of the ball so that the energy level specified in 3.6.1 is achieved. Failure of the sample to comply with the requirements of 3.6.1 shall be cause for rejection of the sample or the lot.

4.7.7.2 Ballistics

4.7.7.2.1 First article. Thirty cartridges of the first article sample shall be grouped and temperature conditioned as described in table III and 4.8. Following removal from the conditioning chamber, the cartridge shall be tested as required by 3.6.2.2. Failure of any cartridge to comply with the requirements of 3.6.2.2 shall be cause for sample rejection.

4.7.7.2.2 Production lot acceptance. A sample size for lot acceptance testing will be in accordance with MIL-STD-105, Table I, General Inspection Level II, and table IIA. The sample will be divided into three groups, temperature conditioned and fired as required by 3.6.2.2. If the sample is not evenly divisible by three, the remaining cartridges will be tested at −60 and 205 °F. Failure of any cartridge to comply with 3.6.2.2 shall be cause for rejection of the lot. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

4.7.7.3 Retest. There shall be no retests.

4.7.7.3.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test cartridge, the original test results shall be discarded and that test reconducted.

4.8 Temperature conditioning. The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.

4.8.1 Reconditioning. The sample must be tested within five minutes after removal from the conditioning chamber. If this time is exceeded, the cartridge shall be reconditioned in accordance with 4.8.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8837884 and 8837885. Level B shall be in accordance with Drawings 8837884 and 9227983.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly is used in cartridge actuated devices for Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.3, and 4.7.6)
c. Whether a first article sample is required (see 3.1, 4.4, and 6.3)
d. Test facility for first article tests (see 4.4.1)
e. Production test activity and lot size (see 4.1 and 4.5.1.2)
f. Any special test conditions (see 4.3)
g. Directions for sampling for classification of defects (see 4.6 and 6.4)
h. Directions for shipping radiographs of primers and of entire lot of cartridge assemblies containing defects (see 4.7.1 and 4.7.4)
i. Directions for sampling for protective finish tests (see 4.7.3 and 6.4)
j. Level of preservation and packing (see 5.1)
k. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Inspection sampling plan and acceptance or rejection criteria. The sampling plan and acceptance or rejection criteria of tables IV and V have been used successfully in procuring, examining, and testing the M91 Impulse Cartridge Assembly. For an individual defect (single characteristic), the criteria in table IV have been used. For total defects within a major or minor group the criteria in table V have been used. The contracting activity is cautioned that any deviation from the values in tables IV and V should be warranted and verifed statistically considering the criticality of the impulse cartridge function.

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TABLE IV. *Sampling plan.*

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 to 300	100% Examination	0	1	0	1
301 to 1300	110	1	2	2	3
1301 to 3200	150	2	3	3	4
3201 to 8000	225	3	4	4	5
8001 to 22,000	300	4	5	5	6

TABLE V. *Total allowable defects per sample size.*

Number of characteristics in class	Sample size			
	110	150	225	300
Major defects				
2 to 3	1	2	3	4
4 to 7	2	3	4	5
8 to 19	3	4	5	7
20 to 43	4	5	8	10
Over 43	6	8	11	14
Minor defects				
2 to 3	2	3	4	5
4 to 7	3	4	5	7
8 to 16	4	5	8	10
17 to 48	6	8	11	14
Over 48	8	11	17	20

6.5 Explosive hazard classification information.

Hazard Classification Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking — Explosive Power Device — Class C — Handle Carefully — Keep Fire Away

6.6 Hazard notice. The cartridge described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.7 Definitions.

6.7.1 Ignition delay. Ignition delay is the time between the firing pin striking the primer or ignition device and the first indication of pressure in the closed bomb.

6.7.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

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6.8 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.9 Changes from previous issue. Asterisks are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Preparing activity
Navy-OS
(Project 1377-N941)

Review activity:
Air Force-70

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-60235B	CARTRIDGE, IMPULSE, M91 ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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